

ALLEN LU ZHAO QUAN

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EDUCATION

Nanyang Technological University (Data Science and Artificial Intelligence) **Aug 2023 – Dec 2027**
Bachelor of Science with Honours

- Cumulative GPA: 4.41 / 5.00 | Second Upper Class Honours (Distinction)
- Relevant Coursework: Data Analytics, Machine Learning, Calculus, Linear Algebra, Probability, Regression Analysis, Business Finance

WORK EXPERIENCE

ING **Jul 2026 – Present**

Intern, Banking Analytics

- Engineered automated data integration workflows using SharePoint, Power Apps and Power Automate, transforming Financial Crime operational records into structured datasets for performance analysis and Power BI reporting, lowering monthly data quality issues by 80%

PhillipCapital **Jan 2026 – Jul 2026**

Intern, Quantitative Risk

- Enhanced current GLM pricing model (Poisson/Gamma) to quantify rating factor relativities for private motor portfolios, exploring alternative models including Tweedie and GAM, Tweedie model achieved a Gini gain of 0.126, outperforming baseline model
- Designed Monte Carlo simulations to model stochastic claim outcomes for 9,253 motor insurance policies under a Tweedie GLM framework, identifying 99th-percentile loss ratio ~4.54% above expected levels, quantifying portfolio tail risk
- Developed SQL pipelines in Snowflake to compute borrower-level credit metrics (utilisation ratios, working-capital exposure, investment limits) and implemented rule-based classification to surface eligibility signals via Tableau dashboards for loan decisioning

Adroit Investment Technologies **May 2025 – Aug 2025**

Intern, Strategy & Business Intelligence

- Researched on emerging AI technologies across various sectors and prepared teasers, industry outlook, statistical reports for prospective clients, achieving 33% increase in client acquisition for the Month of May
- Managed components of financial statements using Power BI including Capital Accounts, Capital Calls, Capital Distribution
- Operated across a full stack of a Learning Management System (LMS) for a children's learning school using Odoo platform, including UI/UX adjustments, business logic workflows, data model configurations. Design shortlisted by CEO to be used as pitching material

PROJECTS

Gunvor Quantitative Trading Challenge 2026 **Sep 2026**

- Developed a Python mean reversion strategy on crude to gasoline refining margin using z-scores of 1.5 and 3.0 standard deviation from rolling-mean for entries and stop-loss respectively, placing 2nd of 7 finalist teams, generating \$347,899 P&L on \$1.04M gross exposure

SEA X Codex Hackathon 2026 (Sea Group X OpenAI) **Jun 2026**

Github Link: <https://github.com/Darkmatter126ms/Project-Naik>

- Designed a Tweedie GLM pricing kernel modelling expected weekly income loss for parametric micro-insurance, using BMKG flood risk scores and district-level rainfall triggers as rating factors across 50 synthetic gig worker profiles, achieving a Gini of 0.46 on holdout
- Deployed Tweedie GLM live within a four-agent pipeline with deterministic pricing. Team shortlisted for top 10% of 1,200+ applicants

Nanyang Fintech Catalyst FinTech Innovator's Hackathon 2026 **Mar 2026**

Github Link: <https://github.com/Darkmatter126ms/Wealth-Wellness-Hub>

- Built a research-calibrated, risk-adaptive financial wellness scoring model using CRRA utility and AHP-based factor weighting to quantify portfolio health across liquidity, diversification and growth, with profile-specific non-linear scoring derived from investor risk tolerance

Data Decomposition using Clustering Analysis **Aug 2025 – Nov 2025**

Github Link: <https://github.com/Darkmatter126ms/Data-Decomposition-using-Clustering-Analysis>

- Applied K-Means, K++-means, Agglomerative and Gaussian Mixed Models to decompose high-dimensional datasets, analysing cluster stability, probabilistic membership, and regime segmentation
- Evaluated model performance including Silhouette Score, Davies–Bouldin Index, Inertia, emphasising model assumptions and limitations

EXTRA-CURRICULAR ACTIVITIES

Nanyang FinTech Catalyst (NFC) **Aug 2025 – Present**

Associate, Quantitative Research

- Spearheaded a team project on pricing strategies by developing a Beta Estimator via OLS regression and constructed a mean–variance portfolio optimiser. Risk-adjusted performance for Max–Sharpe strategies is benchmarked against naïve equal-weight allocation
- Backtested alpha signals on WorldQuant BRAIN using subindustry neutralisation, signal decay, and factor transformations on equity data. Best performing signal on adjusted net income achieved out-of-sample Sharpe of 2.04 with OS/IS ratio of 1.16 across TOP3000 US equities

Investment Banking Club (IBC) **Aug 2023 – Present**

Analyst, Sales & Trading

- Prepared teasers, investment rationale, industry outlook, target lists, pitch decks (FX & Options Related)
- Developed and deliver key presentations of various asset classes. Pitches: FX Trade, Rates Trade, Equities Trade

SKILLS & INTERESTS

Languages: English, Chinese (Native or Bilingual Proficiency), Italian (Limited Working Proficiency)

Skills: Python (libraries), R, SQL, C, C++, Java, JavaScript, HTML, CSS, PyTorch, Spark, Snowflake, Tableau, Bloomberg Terminal, Capital IQ, DCF Modelling, Options Pricing, Microsoft Office (Excel, PowerPoint, Power BI), VBA

Certifications: Solving Ordinary Differential Equations with MATLAB (MathWorks), Vector Calculus for Engineers (HKUST)

Interests: Gym, Volleyball, Basketball, Tennis, Poker & Mahjong, Probabilistic Thinking, Debate, Western Culinary, FX, Defi, Equities

Favourite Books: Thinking in Bets (Annie Duke), One Up on Wall Street (Peter Lynch)